BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, KOHAT

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2022

CHEMISTRY

Past Paper 2022 • Paper Code: KT-IX-22-I

Retyped faithfully from the original board paper — same questions, no changes.

Section "A"

Time Allowed: 20 Minutes | **Maximum Marks:** 12 | **Paper Code:** Code: D

Note: Q.1 Attempt all questions of section-A by filling the correct option i.e. A, B, C or D on the OMR RESPONSE SHEET. Cutting, erasing and over writing will not be awarded any marks.

Qi. The gram molecular mass of HNO_3 is .....

(A) 60 (B) 100

(C) 63 (D) 98

Qii. Which of the following form hydrogen bonds?.

(A) H_2 (B) HBr

(C) C_2H_4 (D) HF

Qiii. Which one of the following ions do not have the electronic configuration of an argon atom?.

(A) K^+ (B) S^{-2}

(C) O^{-2} (D) Ca^{+2}

Qiv. Which of these molecules is not a compound?.

(A) N_2O (B) N_2

(C) NO (D) NO_2

Qv. Mass of an atom is mostly due to its .....

(A) Electrons (B) Protons

(C) neutrons (D) Nucleus

Qvi. Shielding effect is due to ....

(A) Neutron and proton (B) Neutron

(C) Electron (D) proton

Qvii. Ink spreads in water because of .....

(A) Vapour Pressure
(B) Expansion
(C) Compressibility of water
(D) Diffusion

Qviii. Brass is an alloy of .....

(A) Copper and zinc
(B) Silver and copper
(C) Copper and tin
(D) Copper and chromium

Qix. A salt bridge is ..... shaped tube which is filled with electrolytic gel.

(A) 'U'
(B) V
(C) W
(D) X

Qx. Alkali metals are .....

(A) Reducing agents
(B) Oxidant
(C) Oxidizing agents
(D) Dehydrating agents

Qxi. The molarity of NaOH solution by dissolving 4 g of it in 250ml water is ...

(A) 0.4M
(B) 0.8M
(C) 0.2M
(D) 0.1M

Qxii. Oxidation state/number of P in PCl_5 is ....

(A) +5
(B) +6
(C) +3
(D) +4

Section "B"

Time Allowed: 2 Hours 40 Minutes. | **Maximum Marks:** 32

QQ.2. Attempt any EIGHT parts. All parts carry equal marks.

i Define free radical, its formation with examples?

ii Differentiate between atomic number and mass number?.

iii Why an atom is considered as neutral particle ?

iv Place the following elements in order of increasing ionization energy; Na, S, Mg and Ar.

v Why dative bond is always polar?

vi What are crystalline solids?

vii What will be the Molarity of the solution if $30cm^3$ of 2.5 M of $CaCO_3$ solution is diluted to $280cm^3$?.

viii Define oxidation and reduction.

ix How does sodium act as a reducing agent?.

x Draw the Lewis structures of NH_3 and H_2S?.

xi Write down at least four differences between compound and mixture.

Section "C"

Maximum Marks: 21

> Note: Note: Attempt any THREE of the following questions. All questions carry equal marks.

QQ.3.

a. [4] Describe the classification of molecules bases on number of atoms and nature of atoms in detail

b. [3] Why Dalton's atomic theory is considered as a base for modern atomic concepts?

QQ.4.

a. [4] Differentiate between Atomic Radii and Covalent Radii? Explain the trends of atomic radius in the periodic table.

b. [3] Define energy level and sub-energy level?

QQ.5.

a. [4] Differentiate between reducing and Oxidizing agents.

b. [3] Compare electrolytic cell with galvanic cell?

QQ.6.

(a) [4] Explain polar and non polar covalent bonds.

(b) [3] Calculate the moles in 60 g of CO_2.